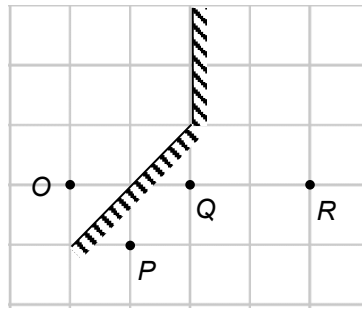


1. Object O is placed in front of two plane mirrors as shown in the following diagram.

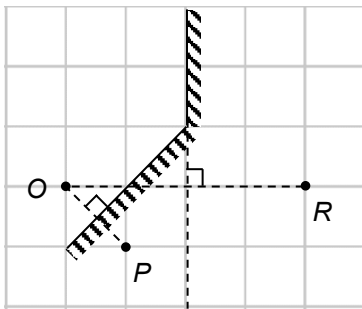


At which position(s) is/are the image(s) of the object formed?

- A At P only
- B At Q only
- C At P and Q only
- D At P and R only

-- ans --

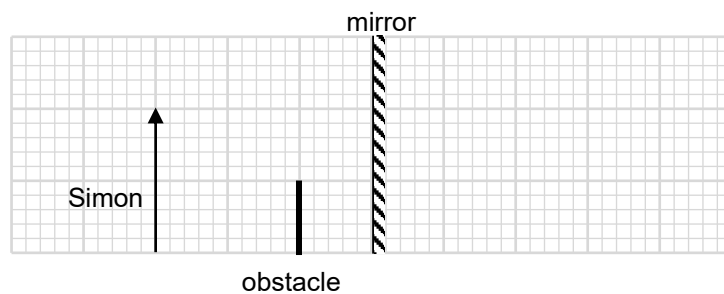
D



Note that the position R of the image can be located by extending the mirror surface.

-- ans end --

2. Simon stands in front of a plane mirror as shown. There is an opaque obstacle between Simon and the mirror.



Which of the following statements is/are correct?

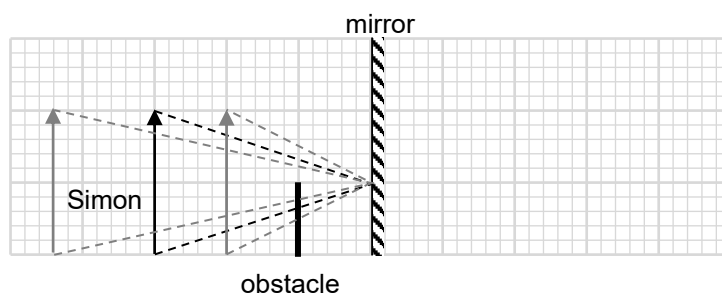
- (1) When he stands at the position shown, he cannot see the image of his shoes in the mirror.
- (2) If he walks closer to the obstacle, he may see the image of his shoes in the mirror.
- (3) If he walks away from the obstacle, he may see the image of his shoes in the mirror.

- A (1) only
- B (2) only
- C (3) only
- D (1) and (2) only

-- ans --

A

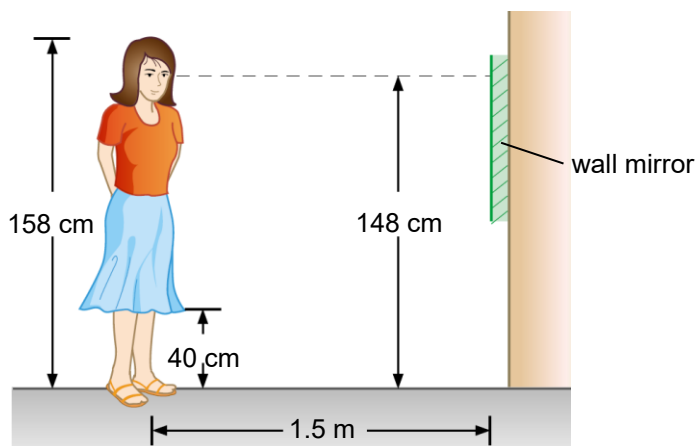
As long as the obstacle is between him and the mirror, light from his shoes is always blocked by the obstacle and cannot reach the mirror and be reflected to his eyes.



∴ (1) is correct. (2) and (3) are incorrect.

-

3. A 158-cm-tall girl stands 1.5 m from a wall mirror as shown below. Her eyes are 148 cm above the floor. The girl can just see the bottom of her skirt, which is 40 cm above the floor, in the mirror. What is the height of the bottom edge of the mirror above the floor?



- A 40 cm
- B 54 cm
- C 80 cm
- D 94 cm

-- ans --

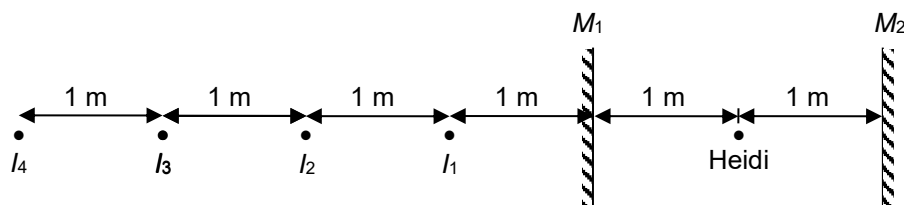
D

Height of her eyes above the bottom of her skirt = $148 - 40 = 108$ cm

Height of the bottom edge of the mirror above the bottom of her skirt = $\frac{108}{2} = 54$ cm

Height of the bottom edge of the mirror above the floor = $40 + 54 = 94$ cm

4. Heidi is standing between two parallel mirrors, M_1 and M_2 , with their reflective surfaces facing each other as shown below. When Heidi looks at M_1 , she sees multiple images in the mirror.



Which of the following is/are the image(s) of Heidi seen in M_1 ?

- A I_1 only
- B I_1 and I_2 only
- C I_1 and I_3 only
- D I_1 , I_3 and I_4 only

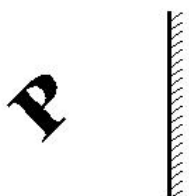
-- ans --

C

I_1 is the image of Heidi formed by mirror M_1 .

There is also an image I' of Heidi 1 m behind M_2 formed by mirror M_2 . The image I' is 3 m away from mirror M_1 . Since image distance is the same as object distance, I_3 is the image of I' .

5. As shown in the figure below, a letter 'P' is placed in front of a plane mirror.



Which of the following correctly shows the image of the letter 'P' as seen in the mirror?

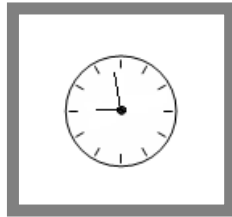
- | | | | |
|---|--|---|--|
| A | | B | |
| C | | D | |

-- ans --

C

An image formed by a plane mirror is laterally inverted.

6. A clock is facing a plane mirror on a wall. The figure below shows the image of the clock in the mirror.



What is the actual time at this instant?

- A 2:58
- B 3:02
- C 8:58
- D 12:13

-- ans --

B

An image formed by a plane mirror is laterally inverted.

The clock looks like this:



7. Which of the following statements about the image formed by a plane mirror is/are correct?

- (1) The image cannot be projected on a screen.
- (2) Light rays entering the eyes do not come from the image.
- (3) The image is behind the mirror.

- A (1) only
- B (1) and (3) only
- C (2) and (3) only
- D (1), (2) and (3)

-- ans --

D

An image formed by a plane mirror is virtual. It cannot be projected on a screen.

∴ (1) is correct.

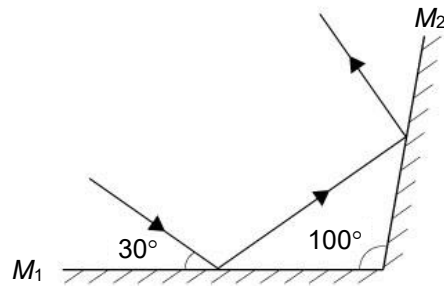
Light rays entering the eyes are emitted or reflected from the object. They do not come from the image.

∴ (2) is correct.

An image formed by a plane mirror is behind the mirror.

(3) is correct.

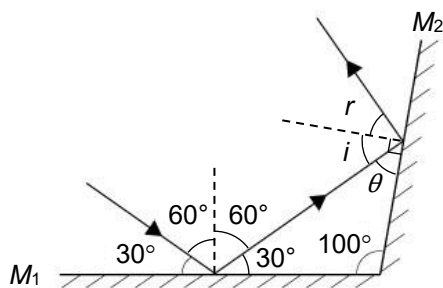
8. In the figure below, what is the angle of reflection of the ray after being reflected by mirror M_2 ?



- A 40°
- B 45°
- C 50°
- D 55°

-- ans --

A



$$\theta + 30^\circ + 100^\circ = 180^\circ \quad (\angle \text{ sum of } \Delta)$$

$$\theta = 50^\circ$$

$$i + \theta = 90^\circ$$

$$i = 40^\circ$$

By the laws of reflection, $r = i = 40^\circ$

The angle of reflection is 40° .

-

9. John is in front of a plane mirror. If the mirror moves away from him at a speed of 2 m s^{-1} , at what speed does his image move away from him?

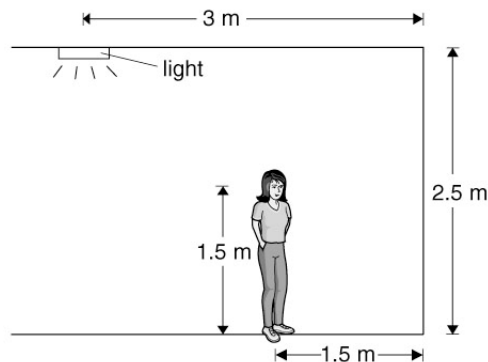
- A 1 m s^{-1}
- B 2 m s^{-1}
- C 4 m s^{-1}
- D 8 m s^{-1}

-- ans --

C

The object distance increases by 2 m in 1 s. Since image distance is the same as the object distance, the image distance also increases by 2 m in 1 s and the image moves away from John by 4 m each second.

10. A girl with her eyes 1.5 m above the ground stands in front of a wall, as shown in the figure below. If a small mirror is to be hung on the wall, how high should it be placed so that the girl can see the light in the mirror?

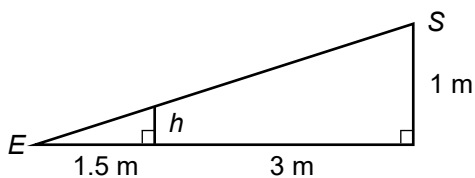


- A 1.50 m above the ground
- B 1.63 m above the ground
- C 1.75 m above the ground
- D 1.83 m above the ground

-- ans --

D

The light is 3 m away from the mirror, so the image of the light is 3 m behind the mirror. Denote the positions of the girl's eye and the image of light as E and S respectively.



By similar triangles,

$$\frac{h}{1.5} = \frac{1}{1.5+3}$$

$$h = 0.33 \text{ m}$$

Distance between the mirror and the ground

$$= 1.5 + 0.33$$

$$= 1.83 \text{ m}$$

The mirror should be placed 1.83 m above the ground.

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